MAT2002 **Name (Print):** _____
Spring 2026
HW4
Deadline: 12th Apr

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1. Verify that the given functions are solutions of the differential equation, and determine their Wronskian.

$$xy''' - y'' = 0; \quad 1, \quad x, \quad x^3$$

Solution:

To verify the solutions, we calculate the required derivatives for each function and substitute them into the differential equation $xy''' - y'' = 0$.

For $y = 1$:

$$y' = 0, \quad y'' = 0, \quad y''' = 0$$

Substituting into the equation: $x(0) - 0 = 0$. This is valid.

For $y = x$:

$$y' = 1, \quad y'' = 0, \quad y''' = 0$$

Substituting into the equation: $x(0) - 0 = 0$. This is valid.

For $y = x^3$:

$$y' = 3x^2, \quad y'' = 6x, \quad y''' = 6$$

Substituting into the equation: $x(6) - 6x = 0$. This is valid.

All three functions satisfy the differential equation.

Wronskian:
The Wronskian W is the determinant of the matrix formed by the functions and their first and second derivatives:

$$W(1, x, x^3) = \begin{vmatrix} 1 & x & x^3 \\ 0 & 1 & 3x^2 \\ 0 & 0 & 6x \end{vmatrix}$$

Since this is an upper triangular matrix, its determinant is simply the product of the entries on its main diagonal:

$$W = 1 \cdot 1 \cdot 6x = 6x$$

2. In each of following, find the general solution of the given differential equation.
(i) $y''' - 5y'' + 3y' + y = 0$
(ii) $y''' + 5y'' + 6y' + 2y = 0$

Solution:

For (i):
The characteristic equation for the differential equation $y''' - 5y'' + 3y' + y = 0$ is:

$$r^3 - 5r^2 + 3r + 1 = 0$$

By inspection, we can see that $r = 1$ is a root $(1 - 5 + 3 + 1 = 0)$. Factoring out $(r - 1)$ yields:

$$(r - 1)(r^2 - 4r - 1) = 0$$

Using the quadratic formula to solve $r^2 - 4r - 1 = 0$:

$$r = \frac{4 \pm \sqrt{16 - 4(1)(-1)}}{2} = \frac{4 \pm \sqrt{20}}{2} = 2 \pm \sqrt{5}$$

We have three distinct real roots: $r_1 = 1$, $r_2 = 2 + \sqrt{5}$, and $r_3 = 2 - \sqrt{5}$.
Thus, the general solution is:

$$y(x) = C_1 e^x + C_2 e^{(2+\sqrt{5})x} + C_3 e^{(2-\sqrt{5})x}$$

For (ii):
The characteristic equation for the differential equation $y''' + 5y'' + 6y' + 2y = 0$ is:

$$r^3 + 5r^2 + 6r + 2 = 0$$

By inspection, $r = -1$ is a root $(-1 + 5 - 6 + 2 = 0)$. Factoring out $(r + 1)$ yields:

$$(r + 1)(r^2 + 4r + 2) = 0$$

Using the quadratic formula to solve $r^2 + 4r + 2 = 0$:

$$r = \frac{-4 \pm \sqrt{16 - 4(1)(2)}}{2} = \frac{-4 \pm \sqrt{8}}{2} = -2 \pm \sqrt{2}$$

We have three distinct real roots: $r_1 = -1$, $r_2 = -2 + \sqrt{2}$, and $r_3 = -2 - \sqrt{2}$.
Thus, the general solution is:

$$y(x) = C_1 e^{-x} + C_2 e^{(-2+\sqrt{2})x} + C_3 e^{(-2-\sqrt{2})x}$$

3. In each of following, determine the general solution of the given differential equation.
(i) $y^{(4)} - 4y'' = t^2 + e^t$
(ii) $y^{(4)} + y''' = \sin 2t$

Solution: For (i):
First, we find the complementary solution $y_c(t)$ for the homogeneous equation $y^{(4)} - 4y'' = 0$.

The characteristic equation is:

$$r^4 - 4r^2 = 0 \implies r^2(r-2)(r+2) = 0$$

The roots are $r = 0$ (multiplicity 2), $r = 2$, and $r = -2$. Thus,

$$y_c(t) = C_1 + C_2 t + C_3 e^{2t} + C_4 e^{-2t}$$

Next, we find a particular solution $Y_p(t)$ for $g(t) = t^2 + e^t$ using the method of undetermined coefficients. We can split this into Y_{p1} for t^2 and Y_{p2} for e^t. For t^2, since $r = 0$ is a double root, we multiply the standard guess by t^2:

$$Y_{p1} = t^2(At^2 + Bt + C) = At^4 + Bt^3 + Ct^2$$

$$Y_{p1}'' = 12At^2 + 6Bt + 2C$$

$$Y_{p1}^{(4)} = 24A$$

Substituting into the left side:

$$24A - 4(12At^2 + 6Bt + 2C) = -48At^2 - 24Bt + (24A - 8C) = t^2$$

Equating coefficients gives: $-48A = 1 \implies A = -1/48$, $-24B = 0 \implies B = 0$, and $24A - 8C = 0 \implies C = -1/16$. So $Y_{p1} = -\frac{1}{48}t^4 - \frac{1}{16}t^2$.

For e^t, we guess $Y_{p2} = De^t$. Substituting into the equation gives:

$$De^t - 4De^t = -3De^t = e^t \implies D = -1/3$$

So $Y_{p2} = -\frac{1}{3}e^t$.

The general solution is $y(t) = y_c(t) + Y_{p1}(t) + Y_{p2}(t)$:

$$y(t) = C_1 + C_2 t + C_3 e^{2t} + C_4 e^{-2t} - \frac{1}{48}t^4 - \frac{1}{16}t^2 - \frac{1}{3}e^t$$

For (ii):
First, we find the complementary solution $y_c(t)$ for $y^{(4)} + y''' = 0$. The characteristic equation is:

$$r^4 + r^3 = 0 \implies r^3(r+1) = 0$$

The roots are $r = 0$ (multiplicity 3) and $r = -1$. Thus,

$$y_c(t) = C_1 + C_2 t + C_3 t^2 + C_4 e^{-t}$$

Next, we find a particular solution $Y_p(t)$ for $g(t) = \sin 2t$. We guess $Y_p(t) = A\cos 2t + B\sin 2t$. Taking derivatives:

$$Y_p''' = 8A\sin 2t - 8B\cos 2t$$

$$Y_p^{(4)} = 16A\cos 2t + 16B\sin 2t$$

Substituting into $y^{(4)} + y''' = \sin 2t$:

$$(16A\cos 2t + 16B\sin 2t) + (8A\sin 2t - 8B\cos 2t) = \sin 2t$$

$$(16A - 8B)\cos 2t + (8A + 16B)\sin 2t = \sin 2t$$

Equating coefficients: $16A - 8B = 0 \implies B = 2A$
$8A + 16B = 1 \implies 8A + 32A = 1 \implies 40A = 1 \implies A = 1/40$
Then $B = 2/40 = 1/20$.
So, $Y_p(t) = \frac{1}{40}\cos 2t + \frac{1}{20}\sin 2t$.

The general solution is $y(t) = y_c(t) + Y_p(t)$:

$$y(t) = C_1 + C_2 t + C_3 t^2 + C_4 e^{-t} + \frac{1}{40}\cos 2t + \frac{1}{20}\sin 2t$$

4. In each of following, find the general solution of the given differential equation. Leave your answer in terms of one or more integrals.
(i) $y''' - y'' + y' - y = \sec t, \quad -\pi/2 < t < \pi/2$
(ii) $y''' - y' = \csc t, \quad 0 < t < \pi$

Solution:

For (i):
First, we find the complementary solution $y_c(t)$ for the homogeneous equation $y''' - y'' + y' - y = 0$. The characteristic equation is:

$$r^3 - r^2 + r - 1 = 0 \implies (r^2 + 1)(r - 1) = 0$$

The roots are $r = 1, \pm i$. Thus, the complementary solution is:

$$y_c(t) = C_1 e^t + C_2\cos t + C_3\sin t$$

The fundamental solutions are $y_1 = e^t$, $y_2 = \cos t$, and $y_3 = \sin t$.

Next, we use the method of variation of parameters to find a particular solution $Y_p(t)$. The Wronskian W of the fundamental solutions is:

$$W = \begin{vmatrix} e^t & \cos t & \sin t \\ e^t & -\sin t & \cos t \\ e^t & -\cos t & -\sin t \end{vmatrix} = 2e^t$$

We compute the determinants W_1, W_2, W_3 by replacing the k-th column with $[0, 0, \sec t]^T$:

$$W_1 = \sec t$$

$$W_2 = -\sec t(e^t\cos t - e^t\sin t) = e^t(\tan t - 1)$$

$$W_3 = \sec t(-e^t \sin t - e^t \cos t) = -e^t(\tan t + 1)$$

The derivatives of the parameters are $u_k' = W_k/W$:

$$u_1' = \frac{1}{2}e^{-t}\sec t, \quad u_2' = \frac{1}{2}(\tan t - 1), \quad u_3' = -\frac{1}{2}(\tan t + 1)$$

Leaving the answer in terms of integrals as requested, the particular solution is $Y_p(t) = y_1u_1 + y_2u_2 + y_3u_3$:

$$Y_p(t) = \frac{1}{2}e^t \int e^{-t}\sec t\, dt + \frac{1}{2}\cos t \int (\tan t - 1)\, dt - \frac{1}{2}\sin t \int (\tan t + 1)\, dt$$

The general solution is $y(t) = y_c(t) + Y_p(t)$:

$$y(t) = C_1e^t + C_2\cos t + C_3\sin t + \frac{1}{2}e^t \int e^{-t}\sec t\, dt + \frac{1}{2}\cos t \int (\tan t - 1)\, dt - \frac{1}{2}\sin t \int (\tan t + 1)\, dt$$

For (ii):

First, we find the complementary solution $y_c(t)$ for $y''' - y' = 0$. The characteristic equation is:

$$r^3 - r = 0 \implies r(r-1)(r+1) = 0$$

The roots are $r = 0, 1, -1$. Thus, the complementary solution is:

$$y_c(t) = C_1 + C_2e^t + C_3e^{-t}$$

The fundamental solutions are $y_1 = 1$, $y_2 = e^t$, and $y_3 = e^{-t}$.

Using variation of parameters, the Wronskian W is:

$$W = \begin{vmatrix} 1 & e^t & e^{-t} \\ 0 & e^t & -e^{-t} \\ 0 & e^t & e^{-t} \end{vmatrix} = 2$$

We compute W_1, W_2, W_3 by replacing the k-th column with $[0, 0, \csc t]^T$:

$$W_1 = -2\csc t$$

$$W_2 = e^{-t}\csc t$$

$$W_3 = e^t \csc t$$

The derivatives of the parameters are $u_k' = W_k/W$:

$$u_1' = -\csc t, \quad u_2' = \frac{1}{2}e^{-t}\csc t, \quad u_3' = \frac{1}{2}e^t \csc t$$

The particular solution is $Y_p(t) = y_1u_1 + y_2u_2 + y_3u_3$:

$$Y_p(t) = -\int \csc t\, dt + \frac{1}{2}e^t \int e^{-t}\csc t\, dt + \frac{1}{2}e^{-t} \int e^t \csc t\, dt$$

The general solution is $y(t) = y_c(t) + Y_p(t)$:

$$y(t) = C_1 + C_2e^t + C_3e^{-t} - \int \csc t\,dt + \frac{1}{2}e^t \int e^{-t}\csc t\,dt + \frac{1}{2}e^{-t}\int e^t \csc t\,dt$$

5. Given that x, x^2, and $1/x$ are solutions of the homogeneous equation corresponding to

$$x^3y''' + x^2y'' - 2xy' + 2y = 2x^4, \quad x > 0,$$

determine a particular solution.

Solution:

To use the method of variation of parameters, we first divide the given differential equation by the leading coefficient x^3 to put it in standard form:

$$y''' + \frac{1}{x}y'' - \frac{2}{x^2}y' + \frac{2}{x^3}y = 2x$$

The non-homogeneous term is $g(x) = 2x$.

The given fundamental solutions are $y_1 = x$, $y_2 = x^2$, $y_3 = x^{-1}$. Let's compute the Wronskian W of these solutions:

$$W = \begin{vmatrix} x & x^2 & x^{-1} \\ 1 & 2x & -x^{-2} \\ 0 & 2 & 2x^{-3} \end{vmatrix}$$

Expanding along the first column:

$$W = x\begin{vmatrix} 2x & -x^{-2} \\ 2 & 2x^{-3} \end{vmatrix} - 1\begin{vmatrix} x^2 & x^{-1} \\ 2 & 2x^{-3} \end{vmatrix}$$

$$W = x[(2x)(2x^{-3}) - (-x^{-2})(2)] - [(x^2)(2x^{-3}) - (x^{-1})(2)]$$

$$W = x[4x^{-2} + 2x^{-2}] - [2x^{-1} - 2x^{-1}] = x(6x^{-2}) - 0 = \frac{6}{x}$$

Next, we compute W_1, W_2, and W_3 by replacing the k-th column of W with $[0, 0, g(x)]^T = [0, 0, 2x]^T$:

$$W_1 = \begin{vmatrix} 0 & x^2 & x^{-1} \\ 0 & 2x & -x^{-2} \\ 2x & 2 & 2x^{-3} \end{vmatrix} = 2x\begin{vmatrix} x^2 & x^{-1} \\ 2x & -x^{-2} \end{vmatrix} = 2x(-1-2) = -6x$$

$$W_2 = \begin{vmatrix} x & 0 & x^{-1} \\ 1 & 0 & -x^{-2} \\ 0 & 2x & 2x^{-3} \end{vmatrix} = -2x\begin{vmatrix} x & x^{-1} \\ 1 & -x^{-2} \end{vmatrix} = -2x(-x^{-1} - x^{-1}) = 4$$

$$W_3 = \begin{vmatrix} x & x^2 & 0 \\ 1 & 2x & 0 \\ 0 & 2 & 2x \end{vmatrix} = 2x \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x(2x^2 - x^2) = 2x^3$$

Now, we find the parameter derivatives $u_k' = \frac{W_k}{W}$:

$$u_1' = \frac{-6x}{6/x} = -x^2 \implies u_1 = \int -x^2\,dx = -\frac{1}{3}x^3$$

$$u_2' = \frac{4}{6/x} = \frac{2}{3}x \implies u_2 = \int \frac{2}{3}x\,dx = \frac{1}{3}x^2$$

$$u_3' = \frac{2x^3}{6/x} = \frac{1}{3}x^4 \implies u_3 = \int \frac{1}{3}x^4\,dx = \frac{1}{15}x^5$$

(We set the integration constants to zero since we only need a single particular solution).

Finally, the particular solution is given by $Y_p(x) = u_1y_1 + u_2y_2 + u_3y_3$:

$$Y_p(x) = \left(-\frac{1}{3}x^3\right)(x) + \left(\frac{1}{3}x^2\right)(x^2) + \left(\frac{1}{15}x^5\right)(x^{-1})$$

$$Y_p(x) = -\frac{1}{3}x^4 + \frac{1}{3}x^4 + \frac{1}{15}x^4 = \frac{1}{15}x^4$$

Thus, a particular solution is $Y_p(x) = \frac{1}{15}x^4$.

6. Transform the given initial value problem into an initial value problem for two first order equations.

$$u'' + p(t)u' + q(t)u = g(t), \quad u(0) = u_0, \quad u'(0) = u_0'$$

Solution:

To transform the second-order equation into a system of two first-order equations, we introduce new variables:

$$x_1(t) = u(t)$$
$$x_2(t) = u'(t)$$

Differentiating these new variables with respect to t gives:

$$x_1' = u' = x_2$$

For x_2', we use the original differential equation and isolate u'':

$$u'' = -p(t)u' - q(t)u + g(t)$$

Substituting our new variables into this equation yields:

$$x_2' = -p(t)x_2 - q(t)x_1 + g(t)$$

The initial conditions must also be expressed in terms of the new variables:

$$x_1(0) = u(0) = u_0$$
$$x_2(0) = u'(0) = u_0'$$

Thus, the corresponding initial value problem for two first-order equations is:

$$x_1' = x_2$$
$$x_2' = -q(t)x_1 - p(t)x_2 + g(t)$$

with initial conditions $x_1(0) = u_0$ and $x_2(0) = u_0'$.

Note: In matrix form, this can be written as:

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -q(t) & -p(t) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ g(t) \end{pmatrix}, \quad \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} u_0 \\ u_0' \end{pmatrix}$$

7. Solve the following systems or the initial value problem:
(i) $\boldsymbol{x}' = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \boldsymbol{x}$,
(ii) $\boldsymbol{x}' = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \boldsymbol{x}$;
(iii) $\boldsymbol{x}' = \begin{bmatrix} 1 & -5 \\ 1 & -3 \end{bmatrix} \boldsymbol{x}, \boldsymbol{x}(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Solution:

For (i):

Let $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$. First, find the eigenvalues by solving $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) - 8 = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0$$

The eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = -1$.

For $\lambda_1 = 5$, find the eigenvector $\mathbf{v}_1$:

$$(A - 5I)\mathbf{v}_1 = \begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -4v_{11} + 2v_{12} = 0$$

Choosing $v_{11} = 1$ gives $v_{12} = 2$, so $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

For $\lambda_2 = -1$, find the eigenvector $\mathbf{v}_2$:

$$(A+I)\mathbf{v}_2 = \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies 2v_{21} + 2v_{22} = 0$$

Choosing $v_{21} = 1$ gives $v_{22} = -1$, so $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

The general solution is:

$$\mathbf{x}(t) = c_1 e^{5t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

For (ii):

Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix}$. The characteristic equation is:

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & -1 \\ 1 & 4-\lambda \end{vmatrix} = (2-\lambda)(4-\lambda) + 1 = \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2 = 0$$

We have a repeated eigenvalue $\lambda = 3$.

Find the eigenvector $\mathbf{v}$:

$$(A - 3I)\mathbf{v} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies v_1 + v_2 = 0$$

Let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Since there is only one linearly independent eigenvector, we find a generalized eigenvector $\mathbf{u}$ such that $(A - 3I)\mathbf{u} = \mathbf{v}$:

$$\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies u_1 + u_2 = -1$$

Let $u_1 = 0$, then $u_2 = -1$, so $\mathbf{u} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$.

The general solution is $\mathbf{x}(t) = c_1 e^{3t}\mathbf{v} + c_2 e^{3t}(t\mathbf{v} + \mathbf{u})$:

$$\mathbf{x}(t) = c_1 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 e^{3t} \left(t \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right) = c_1 e^{3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} t \\ -t-1 \end{bmatrix}$$

For (iii):

Let $A = \begin{bmatrix} 1 & -5 \\ 1 & -3 \end{bmatrix}$. The characteristic equation is:

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -5 \\ 1 & -3-\lambda \end{vmatrix} = (1-\lambda)(-3-\lambda) + 5 = \lambda^2 + 2\lambda + 2 = 0$$

The roots are complex: $\lambda = -1 \pm i$. We use $\lambda = -1 + i$.

Find the complex eigenvector $\mathbf{v}$:

$$(A - (-1+i)I)\mathbf{v} = \begin{bmatrix} 2-i & -5 \\ 1 & -2-i \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the second row, $v_1 + (-2-i)v_2 = 0 \implies v_1 = (2+i)v_2$. Let $v_2 = 1$, so $v_1 = 2+i$.

$$\mathbf{v} = \begin{bmatrix} 2+i \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{a} + i\mathbf{b}$$

Using the formulas for complex roots, the general solution is:

$$\mathbf{x}(t) = c_1 e^{-t}(\mathbf{a}\cos t - \mathbf{b}\sin t) + c_2 e^{-t}(\mathbf{a}\sin t + \mathbf{b}\cos t)$$

$$\mathbf{x}(t) = c_1 e^{-t} \begin{bmatrix} 2\cos t - \sin t \\ \cos t \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 2\sin t + \cos t \\ \sin t \end{bmatrix}$$

Now, apply the initial condition $\mathbf{x}(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$:

$$\mathbf{x}(0) = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2c_1 + c_2 \\ c_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

This gives $c_1 = 1$ and $2(1) + c_2 = 2 \implies c_2 = 0$.

The specific solution for the initial value problem is:

$$\mathbf{x}(t) = e^{-t} \begin{bmatrix} 2\cos t - \sin t \\ \cos t \end{bmatrix}$$